



UNIVERSITY OF CAPE TOWN
School of Economics (Honours in EAFM)

ECO4122Z

**Time: 90 minutes + 15
minutes reading time**

Monday 26 May 2025

Marks: 75

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Please fill in your personal details on the answer books provided. Complete the attendance slip and leave this together with your student card on **THE TOP RIGHT HAND CORNER** of your desk.
2. All cell phones must be switched off for the duration of the test and placed in your bags out of sight underneath your chair. Only non-programmable calculators are permitted for class test 1.
3. Do not open this test paper until instructed to do so. You may not write anything in your answer books before the start of the test or once you have been instructed to stop writing.
4. All students have fifteen (15) minutes reading time. During this time you may perform calculations, highlight and make notes on the test question paper ONLY.
5. From the time the first test paper is distributed to the class and until the last test answer book is collected you may not talk to any other student. This is a standard university policy and anyone contravening this rule will be disqualified.
6. Written answers must be completed in blue or black ink only. The university allows workings to be in pencil although we encourage you to do these in pen and reference them clearly. Workings are important and marks will be awarded for relevant completed workings.
7. No questions will be answered by the invigilators during the class test. Make whatever assumptions you deem appropriate and clearly state these in your answers.
8. No student is to leave the test venue during the first hour and the last 20 minutes of the test.
9. **This test has 7 pages. Including this cover page, and a formula sheet. Please ensure you have all these pages.**

		Total marks
Question One	MCQ (Topics 1-5)	10
Question Two	Topic 2 (Stock Market indices)	17
Question Three	Topic 3: Behavioural Finance	8
Question Four	Topic 5 (EUT)	8
Question Five	Topic 4: Technical Analysis	12
Question Six	Topic 1/5: IPS + Behavioural finance	20
		75 marks (90 mins + 15 mins reading time)

Question One (10 marks: 12 minutes)

1. What does the Absolute Risk Aversion (ARA) coefficient represent in Expected Utility Theory?

(2)

- A. The amount of risk an investor will take as their wealth changes.
- B. The investor's preference for high returns over low returns.
- C. The level of wealth an investor requires to take on a risky investment.
- D. The sensitivity of an investor to risk in monetary terms as their wealth changes.

2. A recent graduate from the finance department at Wits calls you to inform you of a foolproof investment strategy. According to him, you can make risk-adjusted abnormal profits by purchasing every stock that announces a stock split. This type of profitable trading strategy would violate which form(s) of market efficiency?

~ high
SS public
S all

(2)

- A. Weak form only
- B. Semi-strong form only
- C. Weak form and semi-strong form
- D. Semi-Strong form and strong form

3. According to market technicians it is time to sell stock in a head-and-shoulders formation when _____.

(2)

- A. the pattern is predicted to occur
- B. the price index pierces the left shoulder
- C. After the pattern has breached the support level
- D. the price index pierces the right shoulder

4. Which of the following statements about the TRIN indicator is TRUE?

(2)

- A. A TRIN value greater than 1 indicates bullish market sentiment because advancing volume is higher than declining volume.
- B. TRIN is calculated as the ratio of advancing issues to declining issues divided by the ratio of advancing volume to declining volume.
- C. A TRIN value less than 1 suggests bearish sentiment and increasing selling pressure.
- D. TRIN is a momentum oscillator commonly used to identify overbought and oversold conditions.

5. An investor with a high Relative Risk Aversion (RRA) coefficient will likely: **(2)**

- A. Increase their allocation to risky assets as their wealth increases.
- B. Maintain the same allocation to risky assets regardless of wealth.
- C. Decrease their allocation to risky assets as their wealth increases.
- D. Always choose the investment with the highest expected return.

Question Two (17 marks: 25 minutes)

You are given the following data for three stocks (A, B, and C) in a portfolio, including their prices and the number of outstanding shares.

Stock	P_0	P_1	Dividend paid during the year	Number of shares outstanding	Free float
A	50	55	1.25	1 000 000	90%
B	30	28	0.50	500 000	100%
C	100	105	1.75	200 000	85%

Required:

- a) Calculate the value of a float-adjusted, market cap-weighted index of the three securities at the end of year 1. The value of the index in the base year (year 0) is 10 000, and there are no additional shares issued during the year. **[5]**
- b) You want to form an equally weighted, price index of shares A, B and C. The index value in the base year (year 0) is equal to 1000. What is the index value of the equally weighted index at the end of year 1? **[5]**
- c) Briefly discuss the difference between rebalancing, and reconstitution for indexes. **[4]**
- d) Discuss how investing in a market cap weighted index can expose an investor to concentration risk. **[3]**

SOLUTIONS:

a) Sum of market caps = $(50 \times 1\,000\,000 \times 0.9) + (30 \times 500\,000) + (100 \times 200\,000 \times 0.85)$
(1m) =
 77 000 000 **(1)**

$D = 77\,000\,000 / 10\,000 = 7700$ **(1)**

Index₁ (based on price return) = $\frac{(55 \times 1\,000\,000 \times 0.9) + (28 \times 500\,000) + (105 \times 200\,000)}{7250}$ **(1m)** = 10564.94 **(1)**

OR

$$\text{Index}_1 (\text{based on total return}) = \frac{(55+1.25 \times 1000000 \times 0.9) + (28+0.5 \times 500000) + (105+1.75 \times 200000)}{7250} (1m) = 10782.14 (1)$$

b) LCM between 50, 30 and 100 = 300 (0.5)

Therefore $Q^*(A) = 300/50 = 6$ (0.5), $Q^*(B) = 300/30 = 10$ (0.5), and $Q^*(C) = 3$ (0.5)

$$\text{Index value (0)} = 1000 = \frac{(6 \times 50) + (10 \times 30) + (3 \times 100)}{d} (1)$$

$$D = 0.9 (1)$$

$$\text{Therefore index value (1)} = \frac{(6 \times 55) + (28 \times 10) + (105 \times 3)}{7.5} = 1027.78 (1)$$

- c) Index rebalancing would occur when market prices change, the the weightings of the shares differs from an equal weighting – therefore the fund manager will need to purchase and sell shares so that the index goes back to its equal weighting.

Index reconstitution occurs when the constituent shares in an index are changed (either removed, or added). Therefore, the fund manager will need to sell all the shares of a company which is removed from the index, and buy proportionate shares of the company which is getting included.

- d) A market cap weighted index is one where securities are held in proportion of their weightings in the market. This means that the largest firm occupy the highest weightings. It is therefore the case that you will end up overexposed to these large firms, when investing in a market cap weighted index.

Question Three (8 marks)

Sophia is a 35-year-old marketing professional who recently inherited a substantial sum of money. Encouraged by friends and colleagues, she decided to explore investing in the stock market. Sophia reads articles about successful investors and often watches financial news channels that discuss trending stocks. She is particularly drawn to companies with high media coverage and sees them as safe bets, assuming that if everyone is talking about them, they must be good investments. Sophia allocates a significant portion of her portfolio to these stocks, especially in the tech sector, believing their popularity is a sign of strength.

As the market begins to show signs of volatility, Sophia becomes nervous but decides to increase her investments in her chosen stocks, convinced that the market will recover based on the

optimism displayed by analysts in the media. Despite some initial losses, she reassures herself that her decisions are sound because of the positive narratives she follows. She disregards warnings from financial advisors to diversify her portfolio, believing that her selections, based on media trends, are superior.

Required: Identify any two possible behavioral biases which explain Sophia's behavior as outlined above. Your answer should include the name of the bias (1 mark), a short description of the bias (1 mark), and the reason you think it applies to Sophia (2 marks).

3. 1- Herding behaviour
- the tendency of an individual to follow what a group is doing because "if everyone is doing it, it must be right"
 - "if everyone is"

- 2- confirmation bias
- when investors listen/accept info that supports their view + disregard other wise
 - disregards warnings + follows confirmers

- 3- Availability bias
- believing in info that is readily available $\rightarrow$ everywhere
 - high media coverage

4a) $80k^{\frac{1}{2}} = R282,84$

$$E(U) = .4 \times 92k^{\frac{1}{2}} + .6 \times 72k^{\frac{1}{2}} \\ = R282,32 \quad \checkmark$$

$$u(CE) = E(U)$$

$$\sqrt{CE} = 282,32$$

$$CE = 282,32^2 = R79k$$

b) $\sqrt{x} = 282,84 - 282,32$
 $x = R0,05c$

The gamble is not too big + she is willing to pay at least 5c

c) $\sqrt{10} = 3,162 \quad \sqrt{11} = 3,3166 \quad \sqrt{12} = 3,4641$
 $+ 0,1543 \quad \quad \quad + 0,1474$

Utility is increasing @ decreasing rate $\therefore$
 fn is risk averse

fn is concave

2nd derivative is -ve

Question Four (8 marks: 12 minutes)

A consumer has utility $u(x) = \sqrt{x}$ and a base wealth of \$80,000. She is offered a gamble where:

- There is a 40% chance she gains \$12,000 (bringing her wealth to \$92,000), and
- A 60% chance she loses \$8,000 (bringing her wealth to \$72,000).

- a) What is the certainty equivalent of participating in this gamble? (3)
 - b) How much would she be willing to pay to avoid this gamble? (2)
 - c) Based on the consumer's utility function, explain whether this consumer is risk averse, risk neutral, or risk seeking. Justify your answer with reference to the properties of utility functions. (3 marks)
- 

Question Five (12 marks: 30 minutes)

The below two figures contain up to date information on the shares of Apple Incorporated (listed on Nasdaq index, Ticker: AAPL). Using the information in both figures, and any TWO technical indicators, indicate what your recommendation would be for a trader who is considering purchasing the share.

Figure 1:

Note: Orange line represents the 50 day MA, Blue line represents the 200 day MA.



Figure 2



Solution:

Could have discussed any 2 of the indicators (Moving average, RSI, support/resistance lines, price/volume movement, candlestick patterns).

Question Seven (20 marks: 29 minutes)

The Khumalo family consists of Sipho (age 42), a successful entrepreneur, and Thandeka (age 40), a paediatrician. They have three children aged 10, 8, and 6. Sipho recently sold his technology *ability* startup, netting R30 million, and the couple wants to use these proceeds to fund their lifestyle, children's education, and long-term retirement. They have *no debt and high need* a combined annual salary of R2 million. Their savings (excluding the R30 million) total R3 million, primarily invested in large-cap RSA stocks and government bonds. *high ability*

Their primary goals include:

- Funding their children's education, estimated at R500,000 per child annually for the next 12 years, adjusted for 6% annual inflation.
- Maintaining their current lifestyle with an annual pre-tax expense of R2 million, also adjusted for inflation.
- Allocating R500,000 annually over the next five years for international family vacations.
- Building a long-term portfolio to fund a comfortable retirement starting at age 60, while preserving their capital for future generations.

Observations:

- Sipho and Thandeka are financially savvy but exhibit behavioral biases:
 - emo* ○ **Illusion of control bias:** Sipho believes his past entrepreneurial success equips him to outperform financial markets.
 - emo* ○ **Recency bias:** Thandeka is wary of equities due to the recent market volatility.
 - emo* ○ **Overconfidence:** Sipho is inclined to overweight technology stocks, believing he understands the sector better than most.

The couple's mean-variance optimized portfolio recommendation is 60% equities, 30% bonds, and 10% cash. However, their actual preferences often deviate due to their behavioral tendencies.

Required:

- a) Based on the Khumalos' financial situation and objectives, comment on the following inputs into their Investment Policy Statement (IPS):
- Return objectives
 - Risk tolerance (considering their willingness and ability to take risks)
 - Time horizon, liquidity needs, and other constraints.

[10]

- b) Explain how the Khumalos' behavioral biases could impact their asset allocation decisions.

[3]

c) high wealth, low SLR
emotional biases $\therefore$ adapt

- c) Would you moderate or adapt to their biases in constructing their portfolio? Justify your approach. [4]
- d) Considering their biases and financial goals, recommend a practical asset allocation and compare it to the mean-variance optimized portfolio (60% equities, 30% bonds, 10% cash). [3]

• Low SLR/

Formula sheet

$$PR_i = \frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \quad PR_I = \sum_{i=1}^n \omega_i PR_i = \sum_{i=1}^n \omega_i \left(\frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \right)$$

$$TR_i = \frac{P_i - P_{i(t-1)} + Income}{P_{i(t-1)}} \quad w_i^P = \frac{P_i}{\sum_{i=1}^N P_i} \quad r^P = \sum_{i=1}^N w_i^P r_i \quad Index^P = \frac{\sum_{i=1}^N P_i}{d}$$

$$new\ divisor = \frac{sum\ of\ constituents\ before\ split}{index\ value\ before\ split}$$

$$w_i^M = \frac{Q_i P_i}{\sum_{i=1}^N Q_i P_i} \quad \omega_i^{float} = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j P_j Q_j} \quad r^M = \sum_{i=1}^N w_i^M r_i \quad Index^M = \frac{\sum_{i=1}^N Q_i P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi} P_{Bi}}{index_B^M} \quad w_i^E = \frac{1}{N} \quad r^E = \frac{\sum_{i=1}^N r_i}{N} \quad Index^E = \frac{\sum_{i=1}^N Q_i^* P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi}^* P_{Bi}}{Index_B^E} \quad \omega_i = \frac{F_i}{\sum_{j=1}^n F_j}$$

$$E(U) = \sum_w U(W) P(W) \quad ARA = \frac{-U''(W)}{U'(W)} \quad RRA = W \frac{-U''(W)}{U'(W)}$$

$$RP = \frac{1}{2} A(w) \cdot \sigma^2$$

